Borneol to Camphor

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Chem 2081-007

Introduction

In this experiment, borneol was converted into camphor through reflux.

Materials and Methods

The sand bath was plugged in and turned on. A beaker of water was filled and placed on top of the sand bath. Then, 0.3 g borneol, 0.3 mL glacial acetic acid, 1 mL acetone, and a boiling chip were added to a long-neck round-bottom flask. A reflux setup was assembled and placed in the hot water bath. A thermometer was placed in the water bath and observed to ensure it did not surpass 50℃. Next, 6 mL of aqueous sodium hypochloride solution was added dropwise into the flask, swirling the flask after each addition. After the last drop of sodium hypochloride was added, the reaction was left to continue for approximately 15 minutes. After this time had passed, the reflux setup was removed from the water bath and left to cool to room temperature. Then, the contents of the flask were transferred into a centrifuge tube. The solution was washed twice with 2 mL methylene chloride (DCM). The two layers were separated into different flasks, and the bottom layer was added back into the centrifuge tube. Then, the solution was washed with 2 mL saturated sodium bicarbonate. The layers were separated once again, and the bottom layer was added back into the centrifuge tube. Next, the solution was washed with 2 mL of 5% sodium bisulfite and 2 mL water. The layers were separated, and the bottom/organic layer was contained in an Erlenmeyer flask. Approximately 3 scoops of anhydrous sodium sulfate were added to the organic solution. Once the solution had been dried, it was poured onto a pre-weighed watch glass, and placed over the hot water bath. After the DCM had been evaporated from the product, the watch glass with the crystals was weighed. All the products were properly disposed of, and the glassware was cleaned and put away.

Results

Molecular Weight of Borneol (g/mol)	154.25
Molecular Weight of Camphor (g/mol)	152.23
Weight of Watch Glass + Product (g)	30.160
Weight of Watch Glass (g)	29.928
Weight of Product (g)	0.232

Table 1. Weight of borneol, camphor, watch glass, and product.

Calculations

$$\boldsymbol{Weight\ of\ Product}: \mathbf{30.160 - 29.928 = 0.232\ g\ C_{10}H_{16}O}$$

$$\boldsymbol{Theoretical\ yield\ of\ camphor}:$$

$$\mathbf{0.3\ C_{10}H_{18}O \times \frac{1\ mol\ C_{10}H_{18}O}{154.25\ g\ C_{10}H_{18}O} \times \frac{1\ mol\ C_{10}H_{16}O}{1\ mol\ C_{10}H_{18}O} \times \frac{152.23\ g\ C_{10}H_{16}O}{1\ mol\ C_{10}H_{16}O} = 0.296\ g\ C_{10}H_{16}O}$$

$$\boldsymbol{Percent\ yield}: \mathbf{\frac{0.232\ g}{0.296\ g} \times 100 = 78.4\%}$$

Discussion

In this experiment, borneol was converted to camphor. Since camphor is insoluble in water, it was found in the DCM layer, which is on the bottom because DCM has a higher density than water. Since the DCM layer was extracted, and the DCM was evaporated, it can be assumed that the solid product leftover was camphor. Since the percent yield was only 78.4%, the most likely sources of error include insufficient reaction time and evaporation during reflux. To solve these issues, a reflux condenser could be used. This would prevent loss of product through evaporation, and in turn, would allow the reaction to continue longer to ensure sufficient reaction time.

References

Keller L, Lichter J, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Laboratory Manual [lab manual]. Auburn (Al): Auburn University.

Lab Performance

Pag 6 D3

Emma Cleveland 007 402 W

Name: Section: Room: Head:

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ~~○~~ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ~~○~~ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○(1pt) ~~○~~ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ~~○~~ (5pts)

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ~~○~~ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ○ (5pts)

Maintain clean and organized personal space. ○ ○ ○ ○ ~~○~~ (5pts)

GTA notes:

GTA signature: _____ Final Performance Grade: [ ]

Exp. No. 09	Experiment/Subject: Borneol to Camphor	Date: 11/16/23	
Name: Emma Cleveland	Lab Partner:	Locker/ Desk No.	Course & Section No. 007

1. In 10 mL long-neck round bottom flask, add 0.3 g borneol with 0.3 mL glacial acetic acid. Dissolve in 1 mL acetone. Add boiling chip.
2. Assemble reflux setup & place in water bath (40-50°C).
3. Add 6 mL aqueous sodium hypochloride solution (bleach) dropwise through top opening of condenser. Swirl after each addition. Let reaction continue for 15 mins (1½ mL/min)
4. Let cool to room temp & extract camphor w/ 2 portions of 2mL methylene chloride (or DCM) using centrifuge tube
5. Neutralize DCM layers w/ 2mL sodium bicarbonate.
6. Wash organic solution with 2mL 5% sodium bisulfite + 2mL water
7. Dry over anhydrous sodium sulfate
8. Evaporate DCM over water bath ≤ 50°C. Weigh crystals & determine % yield.

1. ✓ 0.3 g borneol, 0.3mL acetic acid, 1 mL acetone
2. ✓
3. ✓
4. ✓ separated layers
5. ✓ separated layers
6. ✓ separated layers
7. Added ~3 scoops drying agent
8. ✓

WG+P: 30.160 g
WG: 29.928 g
0.232g

Borneol = 154.25g/mol ($C_{10}H_{18}O$)
Camphor = 152.23 g/mol ($C_{10}H_{16}O$)

$$0.3\text{g } C_{10}H_{18}O \times \frac{\text{mol}}{154.25\text{g}} \times \frac{1\text{mol}}{1\text{mol}} \times \frac{152.23\text{g}}{1\text{mol } C_{10}H_{16}O} = 0.296\text{g } C_{10}H_{16}O$$

$$\%\text{ yield: } \frac{0.232\text{g } C_{10}H_{16}O}{0.296\text{g}} \times 100 = 78.4\%$$

Signature	Date	Witness/TA	Date